

Alejandro Diaz Fontalva

SOFTWARE ENGINEER

(C#, .NET, ANGULAR, TYPESCRIPT)



ABOUT ME

Software Engineer experienced in C#/.NET development, desktop applications and modern web technologies including Angular and TypeScript.

CONTACT

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SKILLS & TOOLS

Languages

C#, TypeScript, JavaScript, SQL, Python, C/C++, PHP

Frontend

Angular, HTML, CSS

Frameworks & Libraries

.NET, WPF, Windows Forms

Architecture & Patterns

OOP, SOLID, MVVM, Component-Based UI

Tools

Git, Visual Studio, VS Code, SQL Server, Vercel

Domains

Software Development · Web & Desktop Applications · Industrial Systems

LANGUAGES

French - Fluent (Professional)

English - Fluent (Professional)

Spanish - Native

HOBBIES

Game dev. · Travelling · Tennis · Skiing

EDUCATION

Master of Mechatronics Engineering 2014
E.T.S.I. Universidad de Málaga | Málaga

Bachelor of Electronics Engineering 2012
E.T.S.I. Universidad de Málaga | Málaga

TRAINING & CERTIFICATIONS

Object-Oriented Design Patterns in C# 2025
Online Course

Frontend Web Development with JavaScript, Angular and Node.js In progress
Udemy | Online

ASP.NET Core & Blazor Training In progress
Udemy | Online

Complete C# Masterclass 2024
Udemy | Online

TwinCAT Motion & TwinSAFE Trainings 2023-2024
Beckhoff Automation | Yverdon les Bains

TwinCAT 3 HMI & Basis Training 2018
Beckhoff Automation | Yverdon les Bains

LabVIEW Embedded Control and Monitoring 2018
National Instruments | CERN

RELEVANT ENGINEERING EXPERIENCE

Automation Engineer – L.E.S.S. SA, Renens (Aug 2022 – Aug 2024)

- Developed and maintained automation software for production lines using Beckhoff PLCs; designed and implemented HMIs.
- Managed software team (2 engineers) and coordinated version control with Git.
- Contributed to all phases of a new production line design and implementation, resulting in >100 products from 20+ different work orders and automatic production of a new product type.

Industrial IT Engineer – ETEL S.A., Môtiers (Nov 2020 – Jul 2022)

- Developed .NET desktop applications (C#, Windows Forms, WPF) for testing Motion Control systems on production benches.
- Provided technical support, implemented new tests/features, resolved bugs, and managed Git version control.
- Supported >10 tests/day, enabling delivery of 100+ products to clients.

Controls & Automation Eng. – CERN, Geneva (Feb 2018 – Jan 2020)

- Developed front-end and back-end PLC control software (Beckhoff/TwinCAT 2 & 3) for accelerator systems; implemented HMIs.
- Coordinated with electrical technicians and tracked HW orders.
- Co-author of a scientific paper presenting the PLC control system for SPS RF upgrade.

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EARLY ENGINEERING EXPERIENCE

Real-Time Software Engineer – *CERN, Geneva* (Jul 2016 – Jul 2017)

- Developed LabVIEW software for data acquisition and HTS magnet testing; integrated FPGA and RT control software in cRIO systems.
- Performed post-mortem data analysis in Matlab.
- Co-author of scientific paper on HTS dipole magnet testing.

Electronic Engineer – *CERN, Geneva* (Jul 2015 – Jul 2016)

- Developed Arduino-based software for sensor calibration in robotic applications and PHP scripts for database management (Oracle/MySQL).
- Migrated robotic arm communication from native protocol to CANOpen.

Electrical Engineer – *Mijas Town Hall, Málaga* (Dec 2014 – Apr 2015)

- Developed indoor/outdoor lighting and low-voltage electrical installation projects for public buildings and roads.

SOFTWARE & WEB DEV. PROJECTS

Portfolio Website (Angular)

[Angular](#) · [TypeScript](#) · [CSS](#) · [deployed on Vercel](#)

Developed and deployed a responsive personal portfolio using Angular and component-based architecture, implementing routing, UI animations and UX-focused responsive design.

WPF CRUD Application (MVVM)

[C#](#) · [.NET](#) · [WPF](#) · [SQL Server](#)

Built a desktop CRUD application using MVVM architecture, async/await patterns and SQL Server integration with modular UI logic and structured data persistence.

Shiraz Web Projects (In Progress)

[WordPress](#) · [WooCommerce](#) · [Responsive Web Design](#)

Business-oriented web projects focused on responsive layouts, UX structure, deployment workflows and practical website configuration for real-world use cases.

Interactive Systems Prototypes

[C#](#) · [Event-Driven Systems](#) · [Real-Time Interaction Logic](#)

Developed modular interaction systems using state-based behaviors, event-driven architecture and reusable real-time logic patterns.

PUBLICATIONS

- J. Oliveira et al., “The PLC control system for the RF upgrade of the Super Proton Synchrotron”, ICALEPCS19, New York, Oct. 2019. (*co-author*)
- J. van Nugteren et al., “Powering of an HTS dipole insert-magnet operated standalone in helium gas between 5 and 85 K”, IEEE Trans. Appl. Supercond., vol. 27, no. 4, Jun. 2017. (*co-author*)
- “Highly efficient PWM strategy over FPGA”, Electronics Letters, 20th Nov. 2008, Vol. 44, No. 24. (*co-author*)